

CSE310 (75165) Data Structures and Algorithms, Fall 2023

Lectures on Tuesdays and Thursdays, 1:30pm-2:45pm, COOR L1-74

Instructor: Guoliang Xue, Professor, Ph.D., IEEE Fellow

Office: BYENG442

Phone: 480-965-6218

Email: xue@asu.edu

URL: <https://www.public.asu.edu/~gxue1/>

Office Hours: Tuesday and Thursdays, 8:00am-9:00am

TA: Xuanli Lin (xlin54@asu.edu), Office Hours: **TBA**

UGTA: Dac Le (dvle3@asu.edu), Office Hours: **TBA**

UGTA: Thomas Tung (tatung@asu.edu), Office Hours: **TBA**

If you need to make an appointment with the instructor or the TA outside of the posted office hours, you can send an email to get coordination.

CATALOG DESCRIPTION:

Advanced data structures and algorithms, including stacks, queues, trees (B, B+, AVL), and graphs. Searching for graphs, hashing, and sorting.

PREREQUISITES:

Being able to program in C++. These are taught in CSE220 and CSE240.

Being able to perform mathematical proofs. These are taught in MAT243.

TEXT BOOK AND OTHER MATERIALS:

Cormen, T. H., Leiserson, C. E., and Rivest, R. L., C. Stein, *Introduction to Algorithms*, The MIT Press, 4th edition.

COURSE OBJECTIVE AND EXPECTED LEARNING OUTCOME:

Students who complete this course can

1. Define data structures (types) such as heaps, balanced trees, hash tables.
2. Explain how to use a specific data structure in modeling a given problem (e.g. I can explain how to model a dictionary using balanced trees).
3. Identify, construct and clearly define a data structure that is useful for modeling a given problem.
4. State some fundamental algorithms such as merge sort, topological sort, Prim's and Kruskal's algorithm, and algorithmic techniques such as dynamic programming and greedy algorithms.
5. Use a specific algorithmic technique in solving a given problem (e.g. I can write a dynamic program that solves a shortest-path problem).
6. design an algorithm to solve a given problem
7. Define the notions of worst-case/best-case/average-case running times of algorithms.
8. Analyze and compare different asymptotic running times of algorithms.

9. Analyze a given algorithm and determine its asymptotic running time.
10. Combine fundamental data structures and algorithmic techniques in building a complete algorithmic solution to a given problem.
11. Create several algorithmic solutions to a given problem and choose the best one among them according to given requirements on time and space complexity.

COURSE ASSESSMENT PLAN:

50%: Closed-Book Exams (midterm01=15%, midterm02=15%, final=20%)

25%: Programming Projects

20%: Written Assignments

05%: Popup Quizzes

GRADE POLICIES:

A weighted average of 90 or more guarantees a letter grade in the A range (A+, A, A-). A weighted average of 80 or more (but below 90) guarantees a letter grade in the B range (B+, B, B-). A weighted average of 70 or more (but below 80) guarantees a letter grade in the C range (C+, C). A weighted average of 60 or more (but below 70) guarantees a letter grade D. Students with a weighted average of 70 or more can also get the Y grade. A curve may be made at the end of the semester. Such curve should not increase the cut-off point for the letter grade.

ABOUT THIS COURSE:

This course supports the Computer Science program and the Computer Systems Engineering program by providing a background in the analysis and synthesis of algorithms for simple problems (CSa,j/CSEa). Students understand basic and advanced data structures and algorithms and apply knowledge of computing and mathematics to design, implement and evaluate solutions to problems (CSc/CSEc). This course serves as a prerequisite for most advanced courses.

Data Structures and Algorithms is a fundamental course to CSE majors. Students will learn important techniques to store data and to process data efficiently. Data structures and algorithms cannot be separated. On one hand, good data structures help in the design of efficient algorithms. On the other hand, good data structures are discovered during the design of efficient algorithms.

This course requires a lot of work. Depending on the different knowledge levels the students may have, all students may not be required to spend the same amount of time on this course. However, every student is encouraged to preview before the class and review immediately after the class.

Students are encouraged to ask questions in class, and to fully use the office hours of the instructor and the TAs. Students are asked to post questions on Ed Discussion under the appropriate category, and avoid asking questions via email unless the question pertains to confidential matters that cannot be shared with other people in the class. However, the instructor and the TAs may choose to ignore questions on topics well covered in class and from students not attending the lecture(s).

There will be closed book tests and many heavyweight homeworks. Homeworks may involve theoretical contents as well as programming contents. The instructor will also assign suggested readings/exercises after each lecture. Students should work on these before the next lecture day but will not hand in the solutions to the exercises. Very often, solutions to these exercises can be found on the Internet. It is not cheating to get the solutions on the

Internet. Discussion on these exercises is encouraged. However, you will gain only after you can understand the materials.

DETAILED SCHEDULE OF TOPICS:

- ❖ Problems, Instances, Algorithms, Arrays and Linked Lists (1 lecture)
- ❖ Asymptotic Notations (2 lectures)
- ❖ Execution and running time of algorithms (1 lecture)
 - Insertion Sort
- ❖ Recurrence Relations (2 lectures)
- ❖ Divide-and-Conquer, Activation Records, Scopes (2 lectures)
 - Merge Sort, Quicksort
- ❖ Decision Trees and Sorting Lower Bound (2 lectures)
- ❖ **Midterm 01** : To be announced one week ahead of time
- ❖ Heaps and Heapsort, Priority Queues (2 lectures)
- ❖ Disjoint Sets: Amortized Analysis (2 lectures)
- ❖ Graph Algorithms (6 lectures)
 - Depth-First Search
 - Breadth-First Search
 - Shortest Paths
 - Topological Sort
- ❖ **Midterm 02**: To be announced one week ahead of time
- ❖ Binary Search Trees, Red-Black Trees (4 lectures)
- ❖ Introduction to Dynamic Programming (2 lectures)
 - Longest Common Subsequences
 - Matrix Chain Multiplication
- ❖ Hashing (1 lecture)
- ❖ Medians and Selection in Linear Time (2 lectures)
- ❖ Problem Solving: Using the right data structures and algorithms (1 lecture)
- ❖ **Final Exam**: December 7, 12:10pm-2:00pm.

Course Policies:

1. Absences for Classes

Excused absences for classes will be given without penalty to the grade in the case of (1) a university-sanctioned event [ACD 304-02]; (2) religious holidays [ACD 304-04]; a list of religious holidays can be found here <https://eoss.asu.edu/cora/holidays>]; (3) work performed in the line-of-duty according [SSM 201-18]. Students who request an excused absences must follow the policy/procedure guidelines. Excused absences do not relieve students of responsibility for any part of the course work required during the period of absence.

The date and time of the final exam is scheduled by the university and is available before the start of the semester. Students with conflicts with the final exam should notify the instructor during the first week of the semester for possible alternatives. The midterm exams will be during class meeting time. The dates of the midterm exams will be announced at least a week prior to the exam, and can be estimated from the list of topics. Students with conflicts with the exams should contact the instructor at least a week prior for possible alternatives.

Quizzes are posted with relatively short notice. The two lowest-scored quizzes will be dropped from the calculation of the weighted average. As such, no makeups will be provided. However, for situations stated in the first paragraph of this section, the student may be excused from the quiz of the day, provided that the student notifies the instructor the supporting documentation.

Each written assignment will be given at least a week to work on. In general, there will be no makeups. Each programming project will be given at least 20 days to work on. In general, there will be no makeups.

If you have to miss an exam due to ASU official business (e.g. presenting a research paper at a conference, competing as a team member of ASU in an event, etc.), please contact the instructor ahead of time. Under certain circumstances, a student may be given an opportunity to makeup a missed exam. In situations beyond human control, makeups may be considered if the student follows the policies listed in the following.

- **Medical Problems:** *Within two days*, you need to submit a statement with the signature of the doctor and the seal of the hospital saying that you cannot come to class during a particular time.
- **Travel Accident:** *Within one day*, you need to submit a police report stating that you are involved in an accident, or your car broke down while you are traveling to school. Your car does not start at home is not a valid reason.
- **Death of an Immediate Family Member:** If you need to attend the funeral of an immediate family member (defined as parent, spouse, sibling, or child), you need the instructor's *prior approval*. Proof is required.

If you missed an exam/assignment due to your own medical problem, you may request a makeup by submitting an official notice from the hospital within two days of the event. The instructor may consider your request. If you missed an exam/assignment due to your own travel accident, you may request a makeup by submitting a copy of the police report on the day of the event. The instructor may consider your request. If you have to miss an exam/assignment due to the death of an immediate family member (defined as parent, spouse, sibling, or child), you need the instructor's prior approval. Proof is required.

In general, for situations beyond human control, the instructor will consider the request. However, for situations within human control, the instructor will deny the request for a makeup. The following are some examples within human control, and request for a makeup will be denied.

- I have to miss a class/assignment/exam because my parents/friends are visiting me on that day.
- I have to miss a class/assignment/exam because I will have my wedding during those days.
- I have to miss a class/assignment/exam because I have to help my father/mother/friend who needs help.
- I have to miss a class/assignment/exam because I will have already bought my airline ticket to go somewhere.
- I have to miss a class/assignment/exam because my car does not start.

2. Faculty recording of class sessions

Note that class sessions may be recorded, and recordings provided to enrolled students, instructors or instructional support personnel. If you have concerns about being recorded, please contact the course instructor. Recordings

may be used to accommodate student absences. Recordings of all class sessions will be posted in Canvas for all students to access for reviewing course materials.

3. Policy regarding expected student behavior

Students in this class are expected to acknowledge and embrace the FSE student professionalism expectation located at: <https://engineering.asu.edu/professionalism/>

4. Academic Integrity

Students in this class must adhere to ASU's academic integrity policy, which can be found at <https://provost.asu.edu/academic-integrity/policy>). Students are responsible for reviewing this policy and understanding each of the areas in which academic dishonesty can occur. All engineering students are expected to adhere to the ASU Student Honor Code.

All work submitted for the course cannot have been submitted for any other course or any previous section of this same course. Student academic integrity violations are reported to the Fulton Schools of Engineering Academic Integrity Office (AIO). Withdrawing from this course will not absolve you of responsibility for an academic integrity violation and any sanctions that are applied. The AIO maintains a record of all violations and has access to academic integrity violations committed in all other ASU college/schools.

Unless explicitly allowed by your instructor, the use of generative AI tools on any course assignment or exam will be considered academic dishonesty and a violation of the ASU Academic Integrity Policy. Students confirmed to be engaging in non-allowable use of generative AI will be sanctioned according to the academic integrity policy and FSE sanctioning guidelines.

5. Student Copyright Responsibilities

You must refrain from uploading to any course shell, discussion board, or website used by the course instructor or other course forum, material that is not the student's original work, unless the student first complies with all applicable copyright laws; faculty members reserve the right to delete materials on the grounds of suspected copyright infringement.

The contents of this course, including lectures and other instructional materials, are copyrighted materials. Students may not share outside the class, including uploading, selling or distributing course content or notes taken during the conduct of the course. Any recording of class sessions is authorized only for the use of students enrolled in this course during their enrollment in this course. Recordings and excerpts of recordings may not be distributed to others. (see [ACD 304-06](#), "Commercial Note Taking Services" and ABOR Policy [5-308 F.14](#) for more information).

6. Policy against threatening behavior, per the Student Services Manual, [SSM 104-02](#)

Students, faculty, staff, and other individuals do not have an unqualified right of access to university grounds, property, or services (see [SSM 104-02](#)). Interfering with the peaceful conduct of university-related business or activities or remaining on campus grounds after a request to leave may be considered a crime. All incidents and allegations of violent or threatening conduct by an ASU student (whether on- or off-campus) must be reported to the ASU Police Department (ASU PD) and the Office of the Dean of Students.

7. Disability Accommodations

Suitable accommodations are made for students having disabilities. Students needing accommodations must register with the ASU Student Accessibility and Inclusive Learning Services office and provide documentation of that registration to the instructor. Students should communicate the need for an accommodation in enough time for it to be properly arranged. See [ACD 304-08](#) Classroom and Testing Accommodations for Students with Disabilities.

8. Harassment and Sexual Discrimination

Arizona State University is committed to providing an environment free of discrimination, harassment, or retaliation for the entire university community, including all students, faculty members, staff employees, and guests. ASU expressly prohibits discrimination, harassment, and retaliation by employees, students, contractors, or agents of the university based on any protected status: race, color, religion, sex, national origin, age, disability, veteran status, sexual orientation, gender identity, and genetic information.

Title IX is a federal law that provides that no person be excluded on the basis of sex from participation in, be denied benefits of, or be subjected to discrimination under any education program or activity. Both Title IX and university policy make clear that sexual violence and harassment based on sex is prohibited. An individual who believes they have been subjected to sexual violence or harassed on the basis of sex can seek support, including counseling and academic support, from the university. If you or someone you know has been harassed on the basis of sex or sexually assaulted, you can find information and resources at <https://sexualviolenceprevention.asu.edu/faqs>.

As a mandated reporter, I am obligated to report any information I become aware of regarding alleged acts of sexual discrimination, including sexual violence and dating violence. ASU Counseling Services, <https://eoss.asu.edu/counseling> is available if you wish to discuss any concerns confidentially and privately. ASU online students may access 360 Life Services, <https://goto.asuonline.asu.edu/success/online-resources.html>.

9. Photo requirement

Arizona State University requires each enrolled student and university employee to have on file with ASU a current photo that meets ASU's requirements (your "Photo"). ASU uses your Photo to identify you, as necessary, to provide you educational and related services as an enrolled student at ASU. If you do not have an acceptable Photo on file with ASU, or if you do not consent to the use of your Photo, your access to ASU resources, including access to classes (online or in person) may be negatively affected or denied.

10. Update of syllabus

Any information in this syllabus (other than grading and absence policies) may be subject to change with reasonable advance notice.

11. How long students should wait for an absent instructor:

In the event the instructor fails to indicate a time obligation, the time obligation will be 15 minutes for class sessions lasting 90 minutes or less, and 30 minutes for class sessions lasting more than 90 minutes. Students may be directed to wait longer by someone from the academic unit if they know the instructor will arrive shortly.